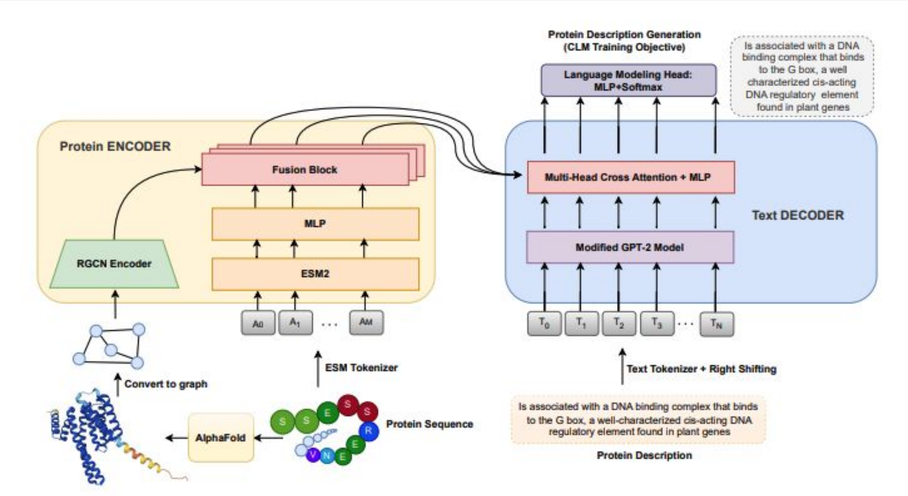
Resource Constrained RAG-Like Approach for Protein Function Descriptions

W266 - Darren Lo & Kamron Mojabe

The Problem:

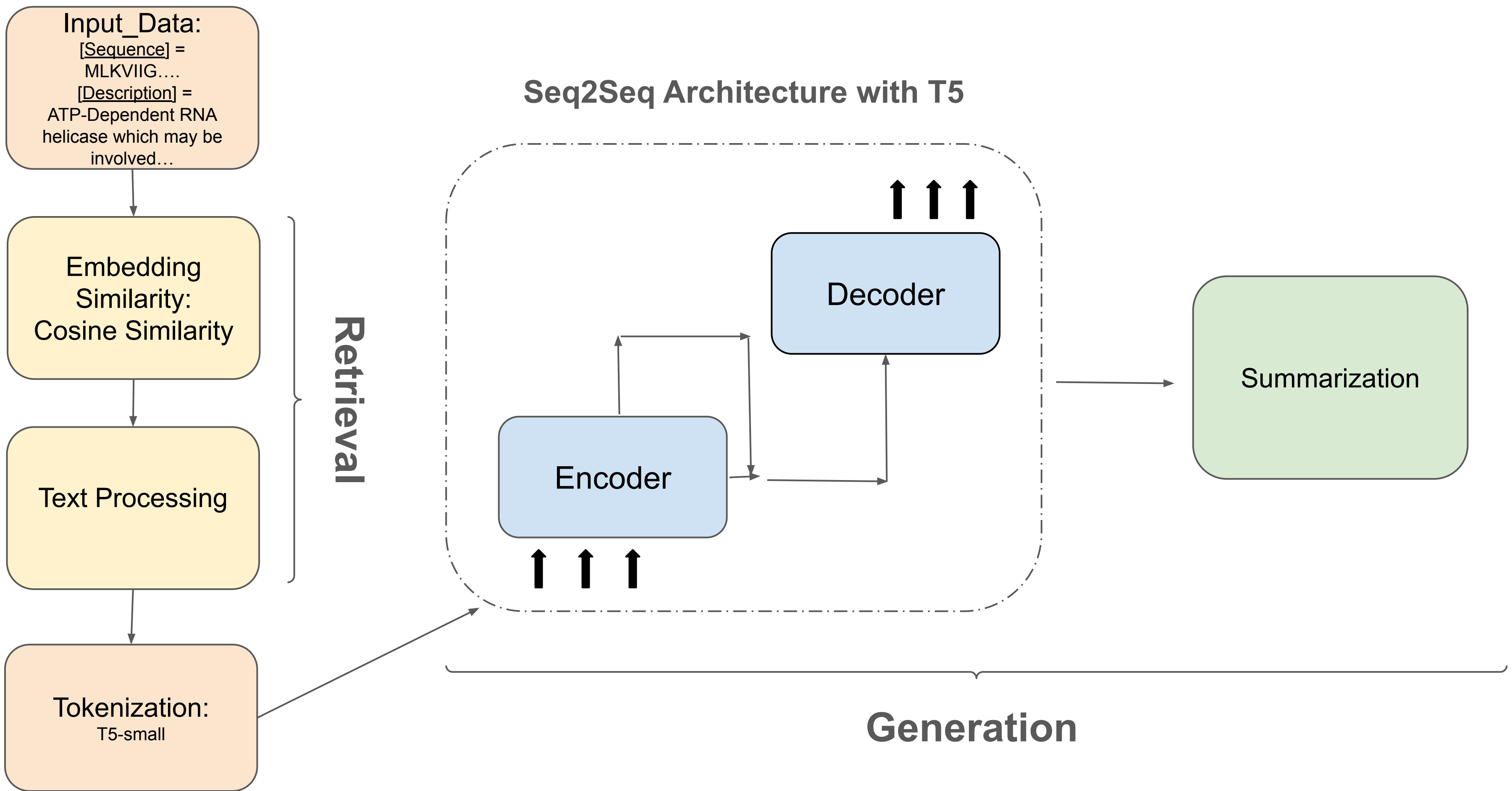


- Requires 64 NVIDIA V100 GPU and took 12 hours to train base model
- 23000 total computing hours used for experimentation

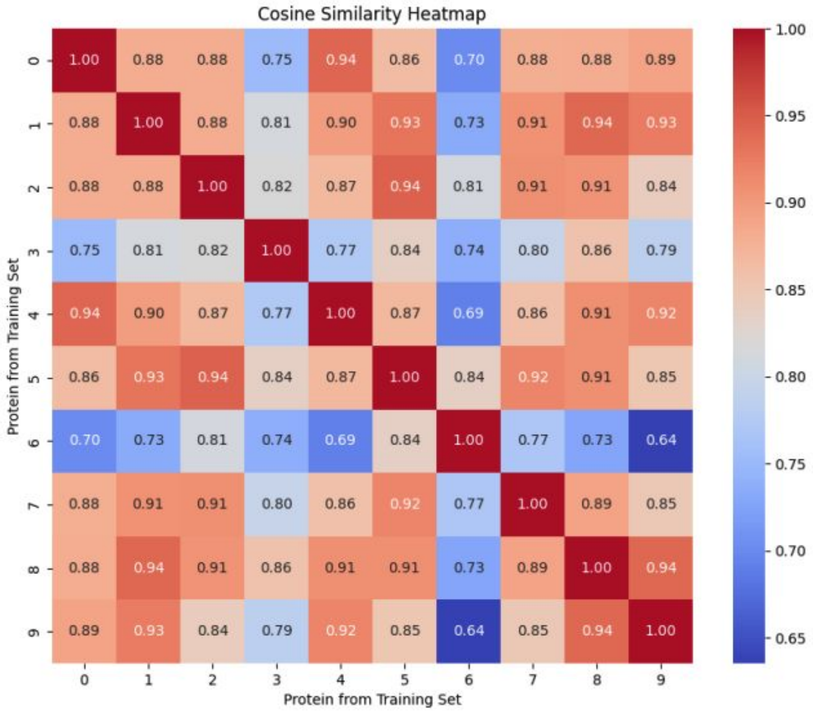
The Approach:

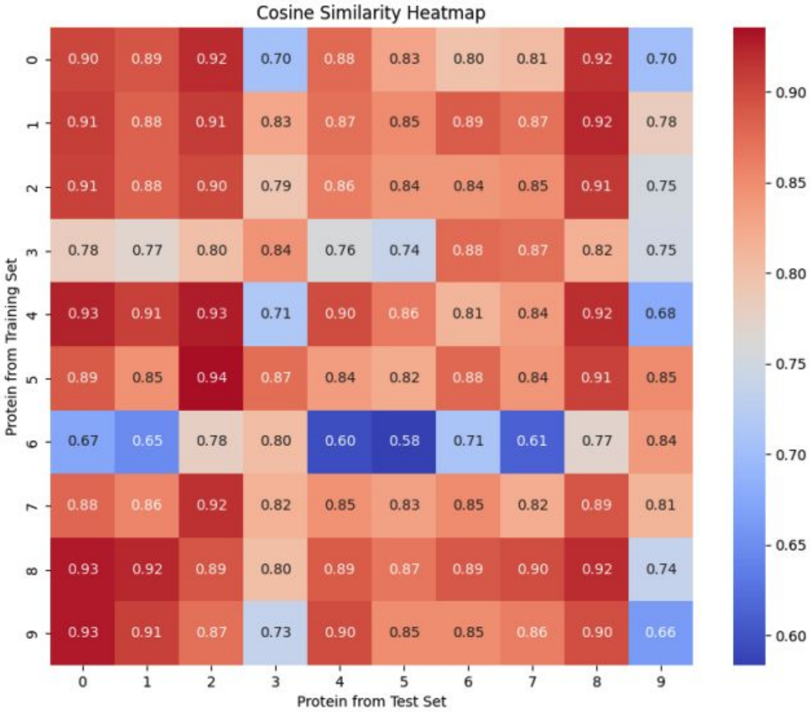
NEXT SLIDE...

Pipeline / Workflow



Cosine Similarity





Results

Model	Rogue-1	Rogue-2	Rogue-L	BLEU
no-retrieval-t5-small	0.123	0.021	0.089	0.031
retrieval-t5-small	0.454	0.285	0.429	4.743

Model	Trainable Parameters	Rouge-1	Rouge-2	Rouge-L	BLEU	Inference Time (sec)
t5-small-FE	41.6M	0.735	0.528	0.692	43.923	968.11
t5-small-HFE	51.1M	0.758	0.580	0.720	51.393	987.23
t5-small	60.5M	0.781	0.620	0.750	56.392	996.58
t5-base-FE	139M	0.843	0.719	0.817	66.579	1543.63
t5-base-HFE	180.4M	0.869	0.769	0.853	73.574	1542.54
t5-base	222.9M	0.890	0.809	0.876	78.015	1558.93

Model	BLEU / Trainable Parameter	BLEU/ Inference Time
t5-small-FE	1.060	0.0454
t5-small-HFE	1.006	0.0521
t5-small	0.932	0.0566
t5-base-FE	0.479	0.0431
t5-base-HFE	0.4078	0.0477
t5-base	0.350	0.0500

Visualization of Summary Results

Model	Original Summary	Predicted Summary
no_retrieval_baseline_t5	Essential component of the MRN complex, a complex that possesses single-stranded DNA endonuclease and 3' to 5' exonuclease activities, and plays a central role in double-strand break (DSB) repair, chromosome morphogenesis, DNA repair and meiosis. In the complex, it mediates the ATP-binding and is probably required to bind DNA ends and hold them in close proximity. Receptor on natural killer (NK) cells and T cells for MHC class I molecules . Upon binding of peptide-free HLA-F open conformer, negatively regulates NK and T cell effector functions . Acts as a receptor on astrocytes for HLA-F. Through interaction with HLA-F, may protect motor neurons from astrocyte-induced toxicity . Chaperone required for protein translocation and folding in the endoplasmic reticulum.	X-rays are a key component of the cytoplasmic synthesis of a systolic acid, a protein that can be absorbed by the neoplasmic acid . It can be used in a variety of ways, such as the iodine, and the apnea . The niodea peptide is a subunit of nasa phosphatase .
retrieval_baseline_t5	In addition to polymerase activity, this DNA polymerase exhibits 3'-5' and 5'-3' exonuclease activity. Involved in quorum quenching (QQ). Inhibits motility and biofilm formation. Could contribute in bacterial competition, as it is capable of hydrolyzing the signaling molecules that mediate interspecies communication. Participates actively in the response to hyperosmotic and heat shock by preventing the aggregation of stress-denatured proteins, in association with DnaK and GrpE. It is the nucleotide exchange factor for DnaK and may function as a thermosensor. Unfolded proteins bind initially to DnaJ.	Zusätzliche polymerase activity dna polymerase. Exonuclease activity involved in quorum quenching qq inhibits motility in biofilm formation. Could contribute to bacterial competition capable of hydrolyzing signaling molecules mediate interspecies communication. Participates actively in the response to hyperosmotic heat shock preventing aggregation proteins association with dnak grpe nucleotide exchange factor dnak. May function as a thermosensor unfold.
t5_small_model	Catalyzes the biosynthesis of sphingolipids with very long-chain fatty acid (VLCFA). Required for the formation of plasmodesmal cytoplasmic sleeve during the transition from type I to type II plasmodesmata to modulate post-sieve elements (SE) unloading and symplastic cell-to-cell molecular trafficking at the phloem pole pericycle (PPP)-endodermis interface in roots.	Catalyzes the biosynthesis of sphingolipids from fatty acid VlcFA. Required for the formation of plasmodesmal and cytoplasmic sleeve transition from type II to plasmodesmata. Can modulate elements of S-adenosylmethionyl-tRNA by unloading the symplastic molecular trafficking of phloem from the pole to the pericycle of the PPP interface roots.

Future Work / Next Steps

Encodings

- ESM2-Variants
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- ProsT5
- Ensemble Approach

Each encoder is trained differently.

Similarity

- PAM
- BLOSUM
- e-score

Different similarity scoring could also affect outcomes.

Generation Model

- BioBERT
- ProLLaMA
- ProtGPT2

Domain specific models could lead to better outputs.